

Alessandro Bruno

Software Engineer — Rust, Backend & Distributed Systems

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PROFILE

Rust-focused software engineer building asynchronous, event-driven backend systems and developer tooling. Experience with Tokio, Valkey, object storage, observability and open-source engineering.

EXPERIENCE

Rust Backend Engineer Jun 2026 – Present
Luna S.r.l. Bologna, Italy

- Design and develop an **event-driven microservices architecture in Rust**, using Tokio for asynchronous workloads and Valkey Streams/Queues as the messaging backbone.
- Build services for real-time data processing and durable persistence on **S3-compatible object storage**, with an emphasis on reliability, concurrency, and clear service boundaries.
- Develop end-to-end observability using **Grafana, Alloy, Loki, Tempo and OpenTelemetry**, while contributing to the architecture for containerized and Kubernetes-based deployment.

Software Developer Intern Sep 2025 – Jun 2026
Luna S.r.l. Bologna, Italy

- Developed a fleet tracking and route-management platform using **Spring Boot, Angular and React Native**, including real-time vehicle progress updates through WebSockets.
- Worked across frontend and backend on production systems, implementing features, improving existing data flows and maintaining legacy Java/Angular applications.

OPEN SOURCE

Zed — JDL Language Support · Merged upstream, PR #3339
Rust · Tree-sitter · Zed Extensions

- Contributed JDL language support to the Zed ecosystem via a Tree-sitter grammar and editor extension, now available in the official marketplace.

ducklake-orm

Rust · DuckDB / DuckLake · Procedural Macros

- Built and published a Rust ORM providing **derive-macro-based type safety, query builders, connection pooling, schema migrations and DuckLake time-travel support**.

SELECTED PROJECTS

Mnemosyne

Rust · Tree-sitter · redb · Content-Addressable Storage · MCP

- Built a local-first developer tool for **semantic source-code history**, tracking symbols and structural changes through Tree-sitter rather than relying exclusively on line-based diffs.
- Designed a modular Rust workspace spanning a background daemon, CLI/TUI tooling, semantic storage and integrations for AI/developer workflows.

Eivar Online

Rust · Bevy · SpacetimeDB · WebAssembly · Distributed State

- Developing a multiplayer prototype with a **server-authoritative Rust simulation**, replicated state, client-side prediction/reconciliation and a SpacetimeDB server module compiled to WebAssembly.

EDUCATION

B.Sc. in Computer Systems Technologies 2026
University of Bologna — Cesena Campus Cesena, Italy

Thesis: Design and Development of PETRA, an Event-Driven Platform for Real-Time Asynchronous Telemetry and Analysis

TECHNICAL SKILLS

Languages	Rust, TypeScript, Java, Python
Backend & Systems	Tokio, Axum, Spring Boot, PostgreSQL, Valkey/Redis, WebSockets, event-driven architectures
Infrastructure	Docker, Kubernetes, S3, Linux, OpenTelemetry, Grafana, Loki, Tempo
Additional	Git, GitHub, Tree-sitter, WebAssembly, Bevy

LANGUAGES

Italian: Native · English: Working proficiency